

TOPIC 12: MANIPULATING DATA

12.1 DATA INTEGRITY

12.1.1 Concept of Data Integrity

- **Definition:** Reliability of a set of data. It is influenced by accuracy and consistency, ensuring data remains free from error and useable from collection through use to archiving.
- **Governing Principle:** Data integrity is essential for effective business decisions and relies on robust data governance.
- **Terminology:**
 - **archive:** Transfer of less frequently used data to a backing store.
 - **data governance:** Systems and procedures for ensuring data is reliable and secure.
 - **validation:** Techniques and processes designed to ensure data is reasonable and follows specific rules.

12.1.2 Maintaining Data Integrity (Key Components)

- **Accuracy:** Data must be correct and free from errors/inconsistencies; it must reflect actual values (e.g., a customer's email address must be correct and functional).
- **Consistency:** Data must follow set rules and formats. All records should use the same units of measurement to avoid conflicting information.
- **Completeness:** Inclusion of all necessary data points; missing data (e.g., missing sales figures) leads to incorrect decisions.
- **Reliability:** Users must be able to trust the data. It must be available when needed through backups, security protocols, and validation.
- **Security:** Measures used to prevent unauthorised access, modification, or deletion (safeguarding from tampering/corruption).

12.1.3 Key Tools and Practices

- **Data validation:** Process of checking data for possible errors/inconsistencies before entry.
- **Data backup:** Regularly creating copies of data to protect it from loss or corruption for restoration after system errors.
- **Data access controls:** Limits on who can access/modify data to prevent unauthorised changes (e.g., physical access limits, OS security, software settings, or encryption).
- **Audit trails:** Tracking any changes made to data, by whom, and when. Typical audits include:

- Timestamp (date/time of action).
- Username of the person performing the action.
- Action performed (created, accessed, modified, deleted).
- Record of the specific data element affected.
- Details of the change source (system/application origin).
- **Secure by Design:** Fundamental principle where programmers consider integrity and implementation during the planning stage of IT systems.

12.1.4 Constructing a Data Dictionary

- **Definition:** A blueprint or plan for a database or dataset. It provides a centralised, agreed plan detailing data structure in a computer system.
- **Metadata:** Data or information that describes other data to provide deeper meaning or understanding.
- **Components of a Data Dictionary:**
 - **Table name:** Unique name of the specific table.
 - **Field name:** Identifier for a specific category of data (e.g., 'email_address').
 - **Description:** Context to avoid ambiguity (e.g., clarifying if a telephone number is landline or mobile).
 - **Data type:** Indicates the type of operations allowed (e.g., integer for math, string for text).
 - **Length/size:** Maximum length of data to be input; helps reduce "rogue data".
 - **Validation rules:** Restrictions to ensure entered data is reasonable.
 - **Constraints:** Additional rules/restrictions (e.g., primary key, not null).
- **Common SQL Data Types (Table 12.1):**
 - **INT:** Whole numbers within a specific range (usually -2147483648 to 2147483647).
 - **SMALLINT:** Smaller integers. Range: -32768 to 32767 or 0 to 65535 .
 - **BIGINT:** For very large positive/negative numbers exceeding INT range.
 - **TINYINT:** Small numbers (0 to 255).
 - **DECIMAL:** Precise decimal numbers with specified precision/scale (e.g., `DECIMAL(10,2)`).
 - **FLOAT and REAL:** Floating-point numbers (approximate values with varying precision).
 - **DATE:** `YYYY-MM-DD`.
 - **TIME:** `HH:MM:SS`.
 - **DATETIME:** Combined date and time.
 - **TIMESTAMP:** Similar to DATETIME, used for tracking record creation/modification.
 - **CHAR:** Fixed-length text string (e.g., `CHAR(8)`).
 - **VARCHAR:** Variable-length string; space-efficient.
 - **TEXT:** For larger blocks of text.
 - **NCHAR/NVARCHAR:** Unicode compatible for multilingual text.
 - **BINARY/VARBINARY:** Binary data (images, audio, video).

- **BOOLEAN:** TRUE or FALSE values.
- **JSON:** Data format used to pass data between systems.

12.1.5 Data Validation

- **Definition:** Implementing rules to attempt to reduce the chance of errors in stored data.
- **Constraints:** Validation does not ensure data is completely error-free, only that it follows pre-set rules.
- **Terminology:**
 - **data migration:** Extracting data from an existing dataset and transferring it to another.
- **Validation Phases (Figure 12.1):**
 - **User Input:** Data is entered into the system.
 - **Check:** System compares data against predefined rules (logical conditions).
 - **Post-check Action:** Logic dictates response. If data fails, an error message is output and the user re-enters. If data passes, the system proceeds.
- **Validation Rules and Rule Logic (Table 12.3):**
 - **Presence check:** Ensures mandatory fields are not blank.
 - Logic: `IF LENGTH(user_input) == 0: RETURN False; ELSE: RETURN True.`
 - **Range check:** Ensures data falls between two accepted values.
 - Logic: `IF value < minimum OR value > maximum: RETURN False; ELSE: RETURN True.`
 - **Lookup:** Checks input against a pre-existing list/table stored outside the data table.
 - **List:** Compares input to a small list of predefined acceptable entries (e.g., "Child", "Adult", "Family").
 - **Length:** Checks that character count meets minimum/maximum requirements.
 - Logic: `IF LENGTH(user_input) > minimum AND LENGTH(user_input) < maximum: RETURN True; ELSE: RETURN False.`
 - **Format check:** Ensures data follows a specific pattern (e.g., DD/MM/YYYY or containing an '@' symbol).
 - **Check digit:** A number added to a string to detect errors (incorrect entry/transposition). Recalculated by the system using an algorithm; if the calculated digit matches the provided one, the entry is likely accurate (e.g., ISBN, credit cards).

12.2 RELATIONAL DATABASES

12.2.1 Data Redundancy

- **Definition:** Unnecessary replication of data in multiple locations within a database.
- **Intentional Duplication:** Sometimes used for backups/availability, but usually refers to unintentional replication.
- **Problems caused by Redundancy:**
 - **Increased storage costs:** Data sets grow exponentially with redundant entries, leading to higher costs.
 - **Data inconsistency:** Updating one instance but not others leads to synchronized errors/confusion.
 - **Maintenance implications:** Managing/updating redundant data wastes time/effort.

12.2.2 Data Normalisation

- **Definition:** The process of breaking down complex data tables into successively smaller/simpler tables to remove redundancy and ensure related data is properly linked.
- **First Normal Form (1NF):**
 - Rule 1: Each cell must contain single, indivisible (**atomic**) values.
 - Rule 2: All entries in a column must be of the same data type.
 - Rule 3: Each column must have a unique name.
 - Rule 4: A **primary key** must be used to identify each record.
- **Second Normal Form (2NF):**
 - Rule 1: Requirements for 1NF must be satisfied.
 - Rule 2: There are no **partial dependencies** (attributes apply to multiple records and are placed in separate tables).
- **Third Normal Form (3NF):**
 - Rule 1: Requirements for 2NF must be satisfied.
 - Rule 2: **Transitive dependencies** have been removed (data in a row that doesn't depend solely on the key is moved to its own table).
- **Key Terminology:**
 - **foreign key:** A column/columns in a table that links to unique data (primary key) in another table.
 - **primary key:** A code/number/name that uniquely identifies a single record in a database.
 - **record:** An individual row in a data table.
 - **atomic data:** Data representing a single attribute that cannot be broken down further.

12.2.3 Logical Data Models

- **Definition:** A plan for how data elements are structured and relate to each other, often represented as an **entity-relationship diagram (ERD)**.

- **Role of DBMS:** A database management system (DBMS) provides tools for creating and manipulating these databases.
 - **Key Components of a Logical Data Model:**
 - **Entities:** Real-world objects/concepts (e.g., customers, products). Represented as single tables. Must be normalised.
 - **Attributes:** Characteristics/properties describing entities (e.g., customer_name). Must be atomic.
 - **Relationships:** Defines how entities connect (one-to-one, one-to-many, many-to-many).
 - **Relationship Notation:** Shown via line types and notations like $(1, n)$ (one-to-many). "Many" sides are often denoted by a three-pronged line.
 - **Unique Identifiers:** Each field needs a unique name to avoid ambiguity. Meaningful names (e.g., order_date) are preferred.
 - **Constraints:** Rules enforced (e.g., not allowing duplicate values) to ensure referential integrity.
-

12.3 BIG DATA

12.3.1 The Five Vs

- **Definition of Big Data:** Very large and complex data sets generated by computer systems that are difficult to manage using traditional data-processing tools.
- **1. Volume:** The sheer amount of data constantly generated (e.g., social media, business transactions).
- **2. Velocity:** The speed at which data is generated (e.g., Facebook generating 4,000,000 'likes' and Twitter 360,000 posts per minute).
- **3. Variety:** Range of formats and structures:
 - **Structured data:** Well-organised, easy to work with (e.g., relational databases).
 - **Unstructured data:** Not organised by typical tables/logical models (e.g., text files, images, videos).
 - **Semi-structured data:** Not traditional relational fashion but contains metadata for processing.
- **4. Veracity:** Quality/accuracy of the data. Uncertainty arises from varied sources (e.g., biased or unregulated social media data).
- **5. Value:** Potential for data to bring significant rewards, such as improved decision-making or customer satisfaction.

12.3.2 Infrastructure and Services

- **Infrastructure:** Hardware and software resources supporting data and processes.
- **IoT (Internet of Things):** Interconnection of everyday objects through the internet.

- **Three Main Groups of Infrastructure/Services:**

1. **Collection:**

- Social media (vast user interactions).
- IoT devices (sensors, wearables, smart appliances).
- Business transactions (e-commerce, financial activities).
- Web/app logs (tracking user behaviour/site visits).
- Surveillance systems (video/image data).
- Machine-generated data (weather/climate monitoring, manufacturing).
- Publicly available data (government/research databases).

2. **Storage:**

- Challenges: Cost, space, scalability.
- Cloud Providers: AWS, Microsoft Azure, Google Cloud Platform (GCP).
- Storage Systems:
 - **Distributed file systems:** Storage across multiple nodes for scalability and fault tolerance.
 - **NoSQL databases:** For unstructured/semi-structured data.
 - **Data warehouses:** Central store for raw data from varied sources, optimised for analytics.

3. **Transmission:**

- Moving/transferring data in and between systems.
- **Bandwidth:** Amount of data transmitted over a connection in a given time.
- Solutions: Fibre optics, data compression, cloud-based storage.

12.3.3 Impact of Storing Big Data

- **Access:** Varies based on business/analysis needs.
- **Data silos:** Data stored across different departments/systems; integration is complex.
- **On-demand access:** Requirement for reliable access whenever needed.
- **Cross-border data access:** Navigating varied legal/regulatory environments across international borders.
- **Processing time:** High-volume calculations require significant power. **Latency** is the delay between requesting and receiving data.
- **Security:**
 - **Data privacy:** Protecting data from unauthorised access; ensuring only those with a valid need can see it.
 - **Data security:** Safeguarding against data breaches and cyber threats using access controls and encryption.

12.3.4 Data Mining, Warehousing, and Analytics

- **Data Mining:** Extracting meaningful information using statistical techniques (clustering, classification, anomaly detection). Often uses machine learning and AI.

- **Data Warehousing:** Collection, storage, and management of raw data in a centralised repository for a unified view.
- **Data Analytics:** Turning mined info into actionable intelligence using tools like models, simulations, and charts.
- **Three Strands of Data Analytics:**
 - **Descriptive analytics:** Summarising historical data to understand "what happened".
 - **Predictive analytics:** Analysing historical patterns to predict future outcomes (causes/effects).
 - **Prescriptive analytics:** Suggesting actions required to optimise results based on predictions.

12.3.5 Use of Big Data in Society

- **Healthcare:** Personalised treatment plans, accelerating drug discovery, tracking public health threats.
 - **Infrastructure/Urban Planning:** Resource management, predictive maintenance of systems, traffic management.
 - **Transportation:** Route optimisation, schedule adjustment based on demand, fuel efficiency monitoring.
 - **Safety/Fraud:** Identifying risk factors, fraud detection (analysing patterns for anomalies), pattern/behavioural recognition.
 - **Real-time monitoring:** Immediate processing and response to fraudulent activities or system performance.
-

CONSOLIDATED EQUATION AND SYMBOL LIST

- **SMALLINT Range:** $[-32768, 32767]$.
- **INT Range:** $[-2147483648, 2147483647]$.
- **Decimal Precision Example:** `DECIMAL(10,2)` (10 digits total, 2 after decimal).
- **Presence Logic:** `IF LENGTH(user_input) == 0: RETURN False.`
- **Range Logic:** `IF value < minimum OR value > maximum: RETURN False.`
- **Length Logic:** `IF LENGTH(user_input) > min AND LENGTH(user_input) < max: RETURN True.`
- **Relationship Notation:** $(1, n)$ signifies "One-to-Many".
- **Big Data stats (2023):**
 - Facebook likes/min: 4,000,000.
 - Twitter posts/min: 360,000.
- **Calculated Field Example:** `Total: [Quantity]*[Item_price].`

ULTRA-CONDENSED FINAL SUMMARY

- **Integrity:** Requires accuracy, consistency, completeness, reliability, and security. Maintained via data governance, validation, and audit trails.
- **Data Dictionary:** Metadata blueprint defining field names, types (INT, VARCHAR, etc.), and constraints.
- **Validation:** Multi-phase check (input, logic check, action). Rules: Presence, Range, Lookup, List, Length, Format, Check Digit.
- **Relational Databases:** Reduce redundancy (replication) to lower costs and prevent inconsistency.
- **Normalisation:** 1NF (Atomic/PK), 2NF (No partial dependencies), 3NF (No transitive dependencies).
- **ERDs:** Map entities (tables), attributes (fields), and relationships (One-to-Many, etc.).
- **Big Data (5 Vs):** Volume, Velocity, Variety (Structured/Unstructured), Veracity, Value.
- **Infrastructure:** IoT collection, Cloud storage (AWS/Azure/GCP), High-bandwidth transmission.
- **Analytics:** Descriptive (past), Predictive (future), Prescriptive (actions). Used in healthcare, urban planning, and fraud detection.

TOPIC 13: ENABLING TECHNOLOGIES

13.1 VIRTUALISATION

13.1.1 What is virtualisation?

- **Definition/Core Concept:** The act of creating a virtual version of something, such as a computer system, server, storage device, or network resource, which behaves like the original.
- **Key Terminology:**
 - *ecosystem*: A set of tools, libraries, and applications designed for a particular purpose or platform (e.g., Apple, Android).
 - *thinclient*: A computer that runs resources stored on a remote server rather than locally installed resources.
- **Conceptual Explanation (Why/How):** Virtualisation allows developers to test software in environments that mimic different target systems (consoles, PCs) without needing physical access to every hardware variation.
- **Types of Virtualisation:**
 - **Desktop virtualisation:** Running multiple operating systems on a single desktop computer (e.g., Windows on an Apple computer); often utilizes *thinclients*.
 - **Storage virtualisation:** Pooling multiple physical storage devices into a single virtual storage pool, simplifying management as a single system.
 - **Network virtualisation:** Creating virtual networks independent of the underlying physical infrastructure, allowing for network isolation (e.g., separating student and administrative resources).
- **Practical Applications:** Used extensively by software/game developers and for administrative management in large organizations.

13.1.2 Benefits of virtualisation

- **Resource utilisation:** Maximizes use of owned physical resources; running tools/services from different *ecosystems* on one platform improves efficiency.
- **Isolation:** Separation between the virtual system and the host platform; issues like malware are restricted to the virtual instance (*sandbox* effect).
- **Flexibility and scalability:** Virtual environments are easier to adapt to changing workloads and requirements compared to physical hardware.
- **Disaster recovery:** Facilitates creation of *snapshots* and backups.
 - *snapshot*: A saved copy of a system, including all data and settings, at a given point in time.
 - **Mechanistic Reasoning:** If a system fails, a *snapshot* can restore it to the exact previous state immediately.

13.1.3 Drawbacks of virtualisation

- **Complexity:** Setting up and managing virtual environments requires specialist software and skilled personnel; maintaining isolation and security is more complex than on host environments.
- **Cost:** While it can save money long-term, initial setup (specialist software, powerful hardware) and additional IT staff training can be expensive.
- **Performance:** Virtual solutions share hardware resources, potentially leading to bottlenecks if too many Virtual Machines (*VMs*) run or if hardware is underpowered/misconfigured.

13.1.4 Achieving virtualisation

- **Governing Theory:** Two main approaches: **containerisation** and **virtual machines**.
 - **Virtual Machines (*VMs*):** Simulate an entire computer system, providing access to a complete operating system.
 - **Containerisation:** Focuses on a narrower scope, providing access to a specific function or application.
 - **Comparison (Table 13.1):**
 - **Abstraction level:** *VMs* use full hardware virtualisation (Guest OS + *hypervisor*); Containerisation operates at the application layer, sharing the host OS kernel.
 - *hypervisor*: A program that runs and manages virtual machines.
 - **Resource usage:** *VMs* consume more resources (memory/CPU) and have larger image sizes; Containerisation has minimal overhead and faster startup.
 - **Isolation:** *VMs* have strong isolation (own OS); Containerisation has lower isolation (shared kernel) and higher security risk if misconfigured.
 - **Portability:** *VMs* are less portable due to *encapsulation* of entire OS; Containerisation is highly portable across environments.
 - *encapsulate*: To group or bundle together related data and/or functions.
 - **Significance:** *VMs* are best for legacy applications and strong isolation; Containerisation is ideal for microservices and cloud-native applications.
-

13.2 DISTRIBUTED SYSTEMS

13.2.1 Concept of Distributed Systems

- **Definition/Core Concept:** A collection of independent computing *nodes* that work together to perform a common goal.
 - *node*: A key point in a network relating to a specific connection or device.

- **Governing Principle:** Distributing nodes across different locations and hardware platforms to provide high levels of performance, scalability, and fault tolerance.
- **Common Tasks/Applications:**
 - **Web services:** Content delivery networks (*CDNs*) allow users to access resources globally; e.g., video streaming from a local server reduces response time.
 - **Data services:** Distributed databases and file systems mirror data across multiple sites to protect against loss and improve access.
 - **Blockchain:** A ledger where transaction records are not stored in one location, reducing single points of failure and tampering risks.

13.2.2 Issues relating to distributed systems

- **Conceptual Explanation (Challenges):**
 - **Failure resistance:** High fault tolerance; if one node fails or is cyber-attacked, others continue to operate and restore the system.
 - **Concurrency:** Multiple tasks/operations occur simultaneously on different nodes; essential for high performance in large organisations.
 - **Replication:** Data is duplicated across nodes; requires systems to ensure data consistency across all instances to avoid errors.
 - **Performance:** Spreading workload reduces lag; however, transmission *latency* is a key issue—physical distance of the node from the user affects speed.
 - *latency*: The delay between requesting and receiving data.
 - **Assumptions:** Adoption requires significant investment in hardware and specialists to maintain complex configurations.
-

13.3 HUMAN COMPUTER INTERACTION (HCI)

13.3.1 Impact on User Experience

- **Definition/Core Concept:** HCI focuses on how users control and interact with computer systems to meet their needs.
- **User Experience (*UX*):** The extent to which users find systems enjoyable, easy, and intuitive.
- **User Interface (*UI*):** The part of the experience the user interacts with.
- **Key Principles of HCI:**
 - **Usability:** Speed, responsiveness, and simplicity of processes.
 - **Consistency:** Similar elements/actions presented the same way across the system to help predict outcomes.
 - **Feedback and communication:** Visual cues, sounds, or *haptic* feedback to maintain clear communication between system and user.

- *haptic*: A form of computer interaction based on touch.
- **Efficiency**: Minimizing steps to complete tasks.
- **Learnability**: Clear instructions and intuitive design for core functionality.
- **Accessibility**: Design for diverse needs (color contrast, font size, screen readers).
- **Flexibility and customisation**: Options for users to adjust the interface to their preferences.
- **Aesthetics**: Appreciation of how pleasing something is to look at; involves *visualhierarchy*.
 - *visualhierarchy*: The order in which elements are noticed and their perceived importance.

13.3.2 Implementing HCI

- **Visual HCI**: Focusing on natural visual perception.
 - **Design elements**: Layout (logical arrangement), Colour (conveying info/identity), Typography (readability/importance), Images/Icons (recognition/engagement).
 - **Responsive design**: Ensuring interfaces adapt to different screen sizes/resolutions.
- **Audio HCI**: Use of sound, speech, and auditory cues.
 - **Speech interfaces**: Voice-controlled systems (Siri, Alexa) using machine learning and speech synthesis.
 - **Auditory feedback**: Notifications (new email), alerts (collision warnings), and sound effects (click sound).
- **Haptic HCI**: Interfaces engaging the sense of touch to create realism and immersion.
 - **Force feedback**: Creating physical sensations of pressure/vibration using motors (e.g., game controllers).
 - **Tactile interaction**: Alternative communication (e.g., tactile displays for Braille).
 - **Biometric sensors**: Physical interaction tracking (e.g., heart rate from smartwatches).

13.3.3 Ergonomic principles of HCI

- **Physical ergonomics**: Focuses on posture, movement, and physical workload to minimize musculoskeletal disorders (*MSDs*).
 - **Posture**: Adjustable furniture and comfortable keyboard/mouse positions.
 - [Diagram: Comparison of incorrect vs correct sitting posture at a computer desk].
 - **Input devices**: Ergonomic keyboards with curved layouts or trackballs to reduce muscle strain.
 - **Visual display**: Adjustable brightness/contrast and resolution; provision of zoom/text size features.
 - **Lighting/Glare**: Computer placement at right angles to windows; use of shades and proper room lighting.

- **Cognitive ergonomics:** Interaction demands on a conscious level.
 - **Workload:** Designing interfaces to minimize cognitive load and fatigue.
 - **Memory:** Using familiar patterns and cues to help users remember tasks.
 - **Attention:** Avoiding unnecessary visual/auditory clutter to reduce distractions.
- **Standardization:** Guidance provided by *ISO9241* regarding ergonomic standards.

13.3.4 Designing an interface to meet needs

- **Menus:** Graphical features providing quick access to tools.
 - **Types (Table 13.2):** Standard horizontal, Hamburger menu (mobile-friendly), Mega menu (subcategories), Vertical sidebar, Contextual menu (right-click), Pie menu (circular).
 - **Icons:** Small pictures acting as shortcuts; provide consistency and accessibility.
 - **Accessibility features:** Adjustable typeface, high-contrast options, screen readers (reading text/tags/metadata), alt text (metadata for images), and captions/transcripts for multimedia.
 - **Windows and Pointers:** Windows provide rectangular borders for programs; Pointers (cursors) show current interaction points (mouse, trackpad, finger).
 - **Guiding Body:** *W3C* (World Wide Web Consortium) provides accessibility guidance.
-

13.4 STORING AND SECURING DATA

13.4.1 Storing data in the cloud

- **Definition:** Resources provided by a third party and accessed via the internet.
- **On-premises vs Cloud:**
 - **On-premises:** Data kept on company-owned hardware in their own buildings.
 - **Cloud:** Data held on servers maintained by providers (AWS, Microsoft Azure, Google Cloud); users pay a fee for access.
 - [Diagram: Diagram comparing internal network storage vs cloud service provider storage].

13.4.2 Securing data

- **File encryption:** Using an *algorithm* to scramble data into an unreadable form (*ciphertext*).
 - *algorithm*: A finite set of predefined steps to solve a problem.
- **Passwords:** Knowledge-based authentication.
 - **Security measures:** Complexity (mixed cases/numbers/symbols) and Length (increases resistance to *brute – forceattacks*).

- *brute – forceattack*: Persistent attempts to gain access by guessing a password.
- **Implementation**: PINs, Pattern locks, and multi-factor authentication.

13.4.3 Security in Database Management Systems (DBMS)

- **Authentication**: Verifying user credentials (usernames/passwords).
 - **Authorisation**: Granting/denying permissions for specific operations (read, write, delete).
 - **Roles**: Defining groups with set privileges (e.g., senior managers).
 - **Privileges**: Types of operations allowed; follows the principle of *leastprivilege*.
 - **Control Methods**:
 - **Access control lists (ACLs)**: Lists of users/roles associated with specific database objects.
 - **Row-level security (RLS)**: Controlling access to individual rows based on conditions.
 - **Views**: Presenting a subset of data to users without exposing the underlying schema.
 - **Auditing**: Logging user activities to provide a trail for investigation.
-

13.5 ENCRYPTION

13.5.1 Symmetric encryption

- **Definition/Core Concept**: Uses the same key to both encrypt and decrypt data.
- **Key Terminology**:
 - *plaintext*: The original version of a message/data.
 - *ciphertext*: The encrypted version of a message/data.
- **Three Stages of Implementation**:
 - **Key generation**: Creating a complex, unique random string of letters/numbers.
 - **Encryption**: $Plaintext + Key \rightarrow Ciphertext$.
 - **Decryption**: $Ciphertext + Key \rightarrow Plaintext$.
- **Benefits**: Highly secure against opportunistic attacks; faster and more efficient than asymmetric encryption for large datasets.
- **Limitations**: Key exchange risk (sharing the key securely); if the key is compromised, all data is vulnerable.
 - [Diagram: Flowchart of Symmetric encryption: Key Gen -> Plaintext + Key -> Algorithm -> Ciphertext -> Algorithm + Key -> Plaintext].

13.5.2 Asymmetric encryption

- **Definition/Core Concept:** Also known as public-key encryption; uses two related keys (Public and Private) that are mathematically linked but different.
- **Governance:**
 - **Public key:** Can be shared widely; used to encrypt messages.
 - **Private key:** Never shared; used to decrypt messages intended for that user.
- **Encryption Logic:**
 - Sender obtains recipient's Public Key.
 - Sender creates *Ciphertext* using recipient's Public Key.
 - Recipient decrypts *Ciphertext* using their unique Private Key.
- **Benefits:** Highly secure for sharing data over insecure channels (internet).
- **Limitations:** Complex mathematical nature; requires significant processing power; not typically used for large quantities of data.
 - [Diagram: Asymmetric process flow between System A and System B showing public/private key pairs and ciphertext transmission].

13.5.3 Certificate-based encryption

- **Definition/Core Concept:** Uses digital certificates to authenticate identity and establish trusted connections.
- **Certificate Authority (CA):** A trusted third party that issues certificates as proof of identity.
- **Certificate Components:**
 1. **Digital certificate:** Document identifying the owner of a public key.
 2. **Issuer:** The trusted *CA* that issued it.
 3. **Validity period:** Timeframe before re-checking is required.
 4. **Digital signature of issuer:** Verifies the certificate was issued by the trusted party.
- *HTTPSprotocol*: Secure version of the protocol used to transfer webpages.
- *SecureSocketLayer(SSL)*: Networking protocol providing secure connections.
- **Process Stages:**
 1. **Certificate exchange:** Browser requests server identification; server sends SSL certificate; browser verifies with *CA*.
 2. **Key exchange:** Browser creates a unique one-time session key (symmetric) encrypted with server's public key.
 3. **Data transmission:** Server and browser transmit data using symmetric encryption.
- **Applications:** HTTPS websites, email encryption, software signing, VPN connections.

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Encryption Logic (Symmetric):**

- $E(P, K) = C$ (Encryption of Plaintext P with Key K yields Ciphertext C)
- $D(C, K) = P$ (Decryption of Ciphertext C with Key K yields Plaintext P)
- **Encryption Logic (Asymmetric):**
 - $E(P, K_{public}) = C$
 - $D(C, K_{private}) = P$
- **Symbolic Definitions:**
 - RTO : Recovery Time Objective
 - RPO : Recovery Point Objective
 - $node_n$: Independent computing unit in a distributed system
 - $VM \neq Container$: (Virtual Machine isolation vs Container shared kernel)

ULTRA-CONDENSED FINAL SUMMARY

- **Virtualisation:** Creates software-based versions of hardware. **VMs** provide full OS simulation (Hypervisor-based); **Containerisation** is lightweight (shared kernel).
- **Distributed Systems:** Multiple **nodes** working towards a goal. Key challenges: *latency*, concurrency, and replication consistency.
- **HCI:** Focuses on **UX** through usability, accessibility, and feedback. **Implementation** involves Visual, Audio, and Haptic channels. **Ergonomics** covers physical posture and cognitive workload (ISO 9241).
- **Securing Data:** **Encryption** transforms plain text to ciphertext. **Passwords** are knowledge-based; DBMS use **ACLs**, **RLS**, and **Views** for access control.
- **Encryption Types:** **Symmetric** is fast (one key); **Asymmetric** is secure for public channels (Public/Private keys). **Certificates** (CAs) provide identity authentication for HTTPS/SSL.

TOPIC 14: USING IT SYSTEMS IN ORGANISATIONS

14.1 ROLE OF IT SYSTEMS

14.1.1 IT systems in organisations

- **Definition/Core Concept:** IT systems are indispensable tools used to monitor production, improve productivity, and provide workers with access to **business-critical** information.
 - **Business-critical:** Data, information, or tasks/operations essential for an organisation to continue to operate.
- **Operational Support Principle:** IT systems ensure day-to-day operations are completed effectively without issue for management, employees, and clients.
- **Conceptual Explanation (Why/How):**
 - **Automation:** Replacing repetitive, simplistic, or laborious tasks (e.g., data entry, order generation) to free employees for strategic work.
 - **Monitoring and Maintenance:** Continuous performance tracking to spot potential issues before they cause downtime or data loss.
 - **Customer Service:** Using self-service portals, chatbots, and knowledge bases to provide 24/7 support.
 - **Data Management and Analysis:** Gaining insights to identify areas for process optimisation and cost reduction.
 - **Resource Management:** Enterprise Resource Planning (ERP) systems integrate finance, HR, and supply chain into a single centralised system.
 - **Collaboration:** Cloud platforms (e.g., Google Workspace, Office 365) enable distributed teams to work simultaneously.
 - **Version control:** Tracking and managing changes in documents/software to prevent data loss or conflicting edits.
 - **Workflow:** A sequence of tasks/processes carried out by groups/individuals to complete an objective.
 - **Communication:** Email, instant messaging, and video conferencing allow global teams to meet flexibly.
 - **Project Management Software:** Tools to plan, organise, track progress, assign responsibilities, and set deadlines.
 - [Diagram: Screenshot of project management software showing a task spreadsheet with columns for Name, Duration, Start, Finish, Predecessors, and a Gantt chart showing task bars on a timeline].
 - **Production and Manufacturing:** Ensuring efficient, consistent, and predictable manufacturing processes.

- **Service Delivery:** Streamlining service delivery through process automation to reach wider audiences and improve workflow.
- **Assumptions/Conditions:** Effective IT systems must be tailored to specific goals like improving efficiency, decision-making, security, and compliance.

14.1.2 Transaction processing

- **Definition/Core Concept: Transaction processing (TP)** refers to the management of individual tasks involving data changes in a system, usually financial or produce-related.
- **Core Theory:** The primary goal of TP is to maintain data integrity, accuracy, and availability.
- **Conceptual Explanation:**
 - **Back-office processes:** Administrative and logistical tasks supporting a business without direct customer interaction.
 - **Process automation:** Using technology to automate complex, repetitive, or mundane tasks.
- **Specific Application: Electronic point of sale (EPOS):**
 - **Sales processing:** Handling cash and contactless/card/smartphone payments.
 - **Inventory management:** Real-time stock tracking; automatic reordering when levels fall below a set threshold.
 - **Customer insights:** Identifying popular items and peak sales times.
 - **Customer experience:** Powering self-checkouts and loyalty cards.
- **Governing Principles: Order Processing (ACID):**
 - **Atomicity:** Transactions cannot be split; treated as one indivisible unit (complete fully or not at all).
 - **Consistency:** System moves from one valid state to another (e.g., stock levels adjust correctly based on the order).
 - **Isolation:** Simultaneous transactions do not interfere with each other (e.g., two customers booking the same seat).
 - **Durability:** Confirmed details/changes are permanent and stored securely, even after unexpected events.
- **Special Cases & Conditions:**
 - **Financial Systems:** Require **High Availability** (24/7 access), **Scalability** (handling volume spikes like holiday sales), and robust **Security/Fraud Prevention**.
 - **Bacs Payment Schemes Limited:** UK organisation managing direct debit and direct credit; requires customer authorisation for bank account deductions.
 - **International Equivalents:** ACH (USA), SEPA (Europe).

14.1.3 Customer relationship management (CRM) systems

- **Definition/Core Concept:** Software tools designed to help businesses manage interactions/relationships with customers and stakeholders.

- **Core Principle:** Aim to provide a positive experience for stakeholders to increase engagement and profits.
- **Conceptual Explanation (How/Why):**
 - **Synchronise marketing events:** Planning and scheduling automated ads/messages (social media, email, SMS) across different mediums.
 - **Loyalty schemes:** Tracking purchases to award points/rewards for future discounts.
 - **Personalisation:** Sending targeted vouchers based on purchase history.
 - [Diagram: Illustration showing CRM systems targeting specific groups or individuals with relevant product information, represented by an icon of an envelope with a star].
 - **Buying trends:** Providing insights to inform strategic decisions on stock levels and advertisement targeting.
 - **Customer service:** Tracking enquiries and issues for satisfactory follow-up.
 - **Customer retention:** Securing return custom through support and incentives.
 - **Upselling:** Identifying patterns to recommend "add-ons" to orders (e.g., recommending a headset for a specific laptop).

14.1.4 Management information systems (MIS)

- **Definition/Core Concept:** Systems/processes providing managers with information to run an organisation and make data-driven decisions.
- **Core Principles:**
 - **Record keeping:** Gathering data from internal sources (EPOS, CRM, operational systems) and external sources (market research, government databases).
 - **Data Processing:** Applying validation, cleansing, and aggregation to make data useful.
- **Conceptual Explanation (Why):**
 - **Complexity Reduction:** MIS removes complexity from structured, semi-structured, and unstructured data to present it accessibly.
 - **Decision-making:** Provides high-quality insights via query tools, dashboards, and reporting interfaces.
 - **Project management:** Provides tools/capabilities that can be integrated with PM software to mitigate risks and establish contingency plans.
- **Practical Significance (Case Study: ESHO SHOES):**
 - MIS identified that production downtime occurred during shift changeovers, allowing specific actions: optimising changeover procedures, additional training, and schedule changes.

14.1.5 Intelligent transportation systems (ITS)

- **Definition/Core Concept:** ITS aim to improve the planning and operation of public and private transportation using sensors, cameras, and monitoring systems.

- **Conceptual Explanation (Functionalities):**
 - **Scheduling and route planning:** Calculating optimal routes using real-time data on traffic, accidents, and weather to reduce travel time and fuel consumption.
 - **Public transport management:**
 - *Dynamic scheduling:* Adjusting bus/train times based on passenger numbers and congestion.
 - *Demand-responsive scheduling:* Creating flexible routes based on real-time passenger demand.
 - *Incident management:* Quick schedule adjustments around accidents.
 - **Timetabling:** Using machine learning and historical data to predict future demand and adjust frequency.
- **Data Sources (Locations):**
 - **User-generated data:** Anonymised speed and location data from mapping apps.
 - **Community-based reporting:** Public reports on road closures/accidents.
 - **Infrastructure:** Road sensors, traffic cameras, and government agency data.
- **Fleet Management:**
 - **Telematics:** An in-vehicle system using GPS and vehicle diagnostics to provide detailed, real-time information.
 - **Significance:** Allows managers to track locations, monitor driver behaviour (e.g., sudden braking), and enable proactive scheduling.
 - [Diagram: Map view showing location-based services with location pins representing fleet tracking].

14.1.6 Expert systems

- **Definition/Core Concept:** An IT system that supports decision-making in a specific area of knowledge (e.g., medicine, technical troubleshooting).
 - **Governing Structure:**
 - **Knowledge base:** A large repository of vast information/expertise related to the specific domain.
 - **Inference engine:** The decision-making part that processes knowledge base info to draw conclusions and recommend solutions.
 - **Conceptual Explanation (How):** Relies on pre-programmed rules and facts encoded by human experts rather than being trained on large datasets.
 - **Limitations & Conditions:**
 - **Narrow scope:** Works only within very strict parameters.
 - **Static:** Knowledge base does not automatically improve through new data (unlike AI/Machine Learning).
-

14.2 OPERATIONAL ISSUES

14.2.1 IT governance and policy

- **Definition/Core Concept:** The pre-defined processes, structures, and mechanisms ensuring IT resources are used appropriately and effectively.
- **Key Goals for IT Governance:**
 - **Strategic alignment:** IT activities must align with business strategies and goals.
 - **Risk management:** Identifying potential risks to IT systems and mitigation methods.
 - **Resource management:** Careful planning of purchase, use, and allocation of resources to maximize value.
 - **Performance measurement:** Establishing measures to assess IT governance effectiveness.
 - **Compliance and legal:** Adherence to protection regulations, privacy laws, and cybersecurity requirements.
- **Policy Strands:**
 - **Business continuity:** Ensuring core functions continue during a data breach or system failure through fallback systems and specific procedures.
 - **Disaster recovery:** Policies for restoring data as quickly and completely as possible after a serious issue (natural disaster, cyber attack).
 - **Risk Management:**
 - **Governing Rule:** $RiskScore = Probability \times Severity$.
 - [Diagram: Risk assessment table with columns: Potential risk, Probability, Severity, Risk score, and Comments/potential mitigation].
 - **User policy (Acceptable Use Policy - AUP):** Defines acceptable behaviours for using IT resources (internet, email, sensitive info) and consequences for failure.

14.2.2 Managing changeover

- **Core Concept:** The transition from an old system to a new one using specific methodologies.
- **Methodologies:**
 1. **Phased changeover:** Implementing the system gradually in stages or modules (e.g., Finance first, then Inventory).
 - **Significance:** Reduces risk as issues are restricted to one area; allows training in smaller groups.
 - **Constraint:** Takes longer to realise full benefits; requires compatibility between systems during transition.
 2. **Direct changeover ("Big Bang"):** Replacing the old system with the new one at a specific cut-off point.
 - **Significance:** Benefits realised immediately; no planning needed for dual system use.
 - **Constraint:** High risk (no fallback); significant productivity drop during the learning curve. Only for small-scale/non-critical applications.
 3. **Parallel changeover:** Running both systems simultaneously for a set period.

- **Significance:** Lowest risk due to full fallback; allows testing and comparison.
- **Constraint:** Increased cost and resource requirements (duplicated workload).
- 4. **Pilot changeover:** Full system implemented within a limited area or group first.
 - **Significance:** Testing in a real-world setting; risks restricted to the pilot area.
 - **Constraint:** Pilot group may not fully reflect the challenges of the whole organisation.

14.2.3 System maintenance

- **Definition/Core Concept:** The ongoing process of managing and updating systems to ensure effectiveness.
- **Types of Maintenance:**
 - **Perfective maintenance:** Identifying and implementing improvements to working systems to increase efficiency or functionality.
 - **Adaptive maintenance:** Modifying systems to accommodate changes in the external environment (new hardware, software, or legal regulations).
 - **Corrective maintenance:** Reactive process of fixing errors, bugs, or defects identified through user feedback or abnormal behaviour.
 - **Mandatory Rule:** Must be followed by **regression testing** to ensure no new issues were introduced.

14.2.4 Archiving data

- **Definition/Core Concept:** Storing historical data no longer in continuous use securely while maintaining accessibility.
- **Reasons/Significance:**
 - **Compliance:** Legal/regulatory requirements to retain records for specific periods.
 - **Business continuity:** Provides additional backups of critical info.
 - **Historical analysis:** Valuable for long-term planning and trend identification.
 - **Auditors:** Provides evidence of data integrity and compliance.
 - **Legacy systems:** Ensuring data remains accessible during migration to new platforms.

14.2.5 Disaster recovery plans

- **Core Concept:** A vital part of IT governance to minimise downtime and maintain business continuity.
- **Key Data Requirements:**
 - **Inventory of data:** List of all critical data and systems needing protection.
 - **Recovery objectives:**

- **Recovery Time Objectives (*RTOs*):** The timeframe in which data must be recovered/restored.
 - **Recovery Point Objectives (*RPOs*):** The maximum amount of data (in time) an organisation is willing to lose (e.g., an *RPO* of one hour means data must be backed up hourly).
 - **Backup and recovery procedures:** Detailed instructions on how/where data is backed up.
 - **Risk Analysis Strands:**
 - **Threat identification:** Assessment of internal (human error) and external (natural disasters, cyber attacks) threats.
 - **Impact assessment:** Evaluating potential impact on data, systems, and operations (linked to *RTOs/RPOs*).
 - **Risk mitigation strategies:** Detailed approach to reducing threat likelihood/impact.
 - **Management Requirements:**
 - **Team roles:** Defining members, roles, and primary delegates for disaster actions.
 - **Communication plan:** Detailed internal/external contact info and plan during a disaster.
 - **Training and drills:** Regular testing to identify potential shortfalls.
 - **Plan maintenance:** Regular reviews to reflect business changes or emerging threats.
 - **Documentation:** Technical documentation must be clear, concise, and stored in a separate system unaffected by a critical event.
-

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Risk Score calculation:** $RiskScore = Probability \times Severity$
 - **Risk Metric Scales:** $Probability \in 1, 2, 3, 4, 5$ (Highly unlikely to Almost certain)
 $Severity \in 1, 2, 3, 4, 5$ (Insignificant to Severe)
 - **Risk Levels:** 1 – 4: Very low 5 – 9: Adequate 10 – 16: High 17 – 25: Severe
 - **Recovery Objectives:** *RTO* (Recovery Time Objective): Target timeframe for restoration. *RPO* (Recovery Point Objective): Target maximum data loss interval.
 - **ACID (Transaction Rules):** *A*: Atomicity *C*: Consistency *I*: Isolation *D*: Durability
-

ULTRA-CONDENSED FINAL SUMMARY

- **IT System Roles:** Automate mundane tasks, enable collaboration (version control/workflow), and manage resources (ERP).

- **Transaction Processing:** Managed by EPOS; governed by **ACID** principles (Atomicity, Consistency, Isolation, Durability) to ensure data integrity.
- **CRM/MIS/ITS:** CRM targets retention and upselling; MIS provides dashboards for managerial decision-making; ITS uses **telematics** and sensors for dynamic scheduling.
- **Expert Systems:** Use a **knowledge base** and **inference engine**; unlike AI, they are static and pre-programmed by human experts.
- **Operational Management:** Governed by **Strategic Alignment** and **Risk Management** ($P \times S$).
- **Changeover:** Four methods—**Phased** (gradual), **Direct** (fast/risky), **Parallel** (safe/expensive), and **Pilot** (localized).
- **Maintenance:** Three types—**Perfective** (improve), **Adaptive** (environmental change), and **Corrective** (fix bugs + regression testing).
- **Disaster Recovery:** Defined by **RTOs** (time to restore) and **RPOs** (backup frequency), requiring regular drills and clear documentation.

TOPIC 15: SYSTEMS DEVELOPMENT

15.1 Project management

15.1.1 Concept of project management

- **Definition/Core Concept:** The process of overseeing and delivering a project by managing human and physical resources to meet specific objectives within a set time, budget, and scope.
- **Governing Theory:** IT project management involves breaking down complex tasks into manageable stages to ensure the final product meets quality standards and deadlines.
- **Key Terminology:**
 - **Project manager:** The individual responsible for overseeing and delivering a project.
 - **Stakeholder:** An individual or group with a particular interest in or link to a project.
 - **Mitigation:** An action or strategy intended to reduce the seriousness or harm caused by a potential risk.
 - **Feature creep:** The continual addition of new features to a product during development that negatively affects deadlines and budgets.
 - **Quality assurance (QA):** The process of ensuring the desired level of performance, design, and reliability is maintained throughout the **project lifecycle**.
 - **Project lifecycle:** The main phases of a project (initiation, planning, execution, monitoring, closure).
- **Conceptual Explanation (Mechanisms of Managing a Project):**
 - **Complexity management:** Breaking down projects involving a wide range of technologies and stakeholders into manageable tasks.
 - **Resource allocation:** Carefully planning the use of tangible assets, including software developers, hardware, time, and budget.
 - **Risk mitigation:** Identifying potential risks (e.g., changing requirements, technical issues, human factors) and implementing contingency plans.
 - **Communication and collaboration:** Effectively conveying project goals, priorities, and expectations to diverse teams (developers, testers, stakeholders).
 - **Change management:** Implementing mechanisms to prioritise changes and incorporate them into project plans to minimise disruption.
 - **Scheduling and deadlines:** Managing "time to market" (speed of delivery) to maintain a competitive edge and identify bottlenecks.
- **Assumptions and Constraints:** Project managers must ensure the project remains as close to the original scope as possible despite stakeholder pressure for changes.

- **Limitations (Wider Issues):** Projects can be affected by factors beyond a manager's control, such as financial market crises, global pandemics, or changes in local legislation/political instability.

15.1.2 Characteristics of a successful IT project

- **Core Theory:** A successful project is defined by its ability to meet agreed objectives on time and within budget through structured engagement and planning.
- **Internal Components (Mechanisms of Success):**
 - **Clear objectives:** Defining what should be achieved using **SMART targets** to help all stakeholders identify requirements.
 - **Effective stakeholder engagement:** Involving individuals or groups affected by the project's success or failure.
 - *Internal stakeholders:* Members of the organisation directly involved (e.g., development team, managers, executives).
 - *External stakeholders:* Those not directly employed but with an interest (e.g., customers, clients, local community, government agencies).
 - **Detailed planning:** Comprehensively outlining scope, deliverables, timelines, resource requirements, risks, and quality standards.
 - **Skilled team:** Allocating tasks to members who possess the specific expertise and experience required.
 - **Adaptability:** Maintaining flexibility to adjust to evolving requirements, technologies, or market conditions without ignoring main objectives.
 - **Effective communication:** Establishing clear channels among team members, stakeholders, and project leadership to avoid/resolve issues.
 - **Quality focus:** Defining quality standards in consultation with stakeholders to deliver high-quality solutions.
 - **Risk management:** Developing plans to identify, analyse, prioritise, and respond to risks.
 - **Continuous improvement:** Carrying out reviews to identify areas of success and improvement to apply best practices to future projects.
 - [Diagram: Development Life Cycle showing a circular flow: Planning -> Analysis -> Design -> Implement -> Test -> Evaluation -> Release -> Support -> (back to Planning)].
 - **On-time and on-budget:** Tracking progress against milestones and monitoring budget spend to take corrective actions.

15.1.3 Risk analysis and SMART targets

- **Risk Analysis Principle:** Using a numbered risk matrix (e.g., 3×3 or 5×5) to identify potential risks and assign scores.
- **Risk Equation:**
 - $RiskScore = Probability \times Impact$.
- **Variables and Scales (5x5 example):**

- **Probability (P):** 1 (Highly unlikely), 2 (Unlikely), 3 (Possible), 4 (Likely), 5 (Almost certain).
 - **Impact (I):** 1 (Insignificant), 2 (Minor), 3 (Significant), 4 (Major), 5 (Severe).
- **Special Case: Level of Risk Interpretation:**
 - *RiskScore* ∈: Very low.
 - *RiskScore* ∈: Adequate.
 - *RiskScore* ∈: High.
 - *RiskScore* ∈: Severe.
- **SMART Targets Definition:** An approach to setting goals that ensures they can be used effectively to monitor and evaluate a project.
- **The SMART Acronym:**
 - **Specific:** Goals must be clear and well-defined for easy tracking.
 - **Measurable:** There must be a way to quantify progress or success.
 - **Achievable:** Goals must be attainable; large goals should be broken into manageable sub-tasks.
 - **Relevant:** Goals must align with overall project aims and be clearly linked to stakeholder needs.
 - **Time-bound:** Goals must have clear and appropriate deadlines to avoid a lack of urgency or team burnout.

15.1.4 Functional and non-functional requirements

- **Functional Requirements Definition:** Exactly what the user(s) should be able to do in the IT system (specific functions and features).
 - *Examples:* User authentication (login with username/password), inventory tracking, generating reports.
- **Non-functional Requirements Definition:** How the system or product should perform (quality attributes, constraints, and characteristics).
 - *Examples:*
 - **Performance:** System must respond to input within 2 seconds.
 - **Security:** Must use public key encryption.
 - **Usability:** Interface must be intuitive and accessible to users with disabilities.
 - **Reliability:** Availability of 99.9% during business hours.
 - **Scalability:** Support spikes in concurrent users without performance loss.

15.1.5 Critical path analysis (CPA)

- **Definition/Core Concept:** A method used to identify the sequence of tasks that must be completed on time to ensure the entire project meets the specified deadline.
- **Conceptual Explanation (Why/How):**
 - Identifies critical tasks (those with zero slack) that cannot be delayed without delaying the whole project.

- Allows project managers to allocate resources and "slack" (*buffer time*) to non-critical tasks.
- **Mechanistic Steps:**
 - Identify all tasks.
 - Estimate duration for each task.
 - Identify dependencies between tasks.
 - Present tasks in a diagram (e.g., Gantt chart, arrow diagram).
 - Determine the longest sequence of dependent tasks (the Critical Path).
- **Significance of Slack:**
 - **Slack:** The difference between the Latest Start (LS) and Early Start (ES). Indicates potential flexibility.
 - Tasks on the critical path have $Slack = 0$.
- [Diagram: A network diagram showing tasks as boxes with start/finish dates and duration, connected by arrows. The "Critical Path" is highlighted with red arrows].

15.1.6 Precedence tables

- **Definition/Core Concept:** A visual tool for scheduling and sequencing tasks, showing task durations, dependencies, and timing constraints.
 - **Components and Calculations:**
 - **Precedence:** Indicates which tasks must be completed before others can start (dependencies).
 - **Duration (D):** Estimated time needed (e.g., in days or hours).
 - **Early Start (ES):** The earliest possible date a task can begin, considering predecessors.
 - **Early Finish (EF):** The earliest possible finish date.
 - $EF = ES + D$.
 - **Late Start (LS):** The latest possible start date without delaying project completion. It is equal to the LS of the preceding task (lowest value used if multiple).
 - **Late Finish (LF):** The latest possible finish date.
 - $LF = LS + D$.
 - **Slack Equation:**
 - $Slack = LS - ES$.
 - **Assumptions:** Effectiveness depends on accurate duration estimates and effective communication with senior developers/engineers.
-

15.2 Waterfall project methodology

15.2.1 Features and concept of Waterfall

- **Definition/Core Concept:** A sequential, linear approach to development where the project is divided into distinct, self-contained phases.
- **Governing Principle:** Each phase must be completed before moving to the next; there is little room for change in project scope once development begins.
- **Suitability (Conditions):** Best for small or simple projects with few requirements and clearly defined, unlikely-to-change objectives.
- [Diagram: Waterfall methodology showing a descending "staircase" of phases: Requirements gathering/analysis -> System design -> Implementation -> Testing -> Deployment -> Maintenance].

15.2.2 Phases and activities of the Waterfall method

1. **Requirements gathering/analysis:**
 - **Identification of stakeholders:** Consulting management and potential users on functionality.
 - **Methods for gathering (Table 15.4):**
 - **Interview:** Direct interaction; picks up non-verbal cues. *Constraint:* Time-consuming.
 - **Observation:** Watching stakeholders in real-life environments. *Constraint:* Obtrusive/disruptive.
 - **Document analysis:** Reviewing existing reports/manuals for context. *Constraint:* Documentation may be outdated.
 - **Focus group:** Collaborative discussions. *Constraint:* Can be hijacked by dominant personalities.
 - **Questionnaire:** Structured feedback from large groups. *Constraint:* Low response rates.
2. **System design:** Expanding requirements into detailed technical and visual designs (e.g., flowcharts for UX/UI).
3. **Implementation phase:**
 - **Coding:** Translating designs into working code.
 - **Infrastructure building:** Combining hardware components/peripherals.
 - **Database creation:** Building and populating data structures.
 - **Interface creation:** Implementing user interaction methods (e.g., responsive layouts).
 - **Documentation:** Producing user guides and technical documentation.
4. **Testing/debugging phase:**
 - **Unit testing:** Testing individual modules/units.
 - **Integration testing:** Testing how units communicate/share data when combined.
 - **System testing:** Testing the entire system against requirements (often by a dedicated QA team).
 - **Acceptance testing:** Final confirmation by stakeholders that the system is ready for deployment.
5. **Installation and Deployment:** Migrating the system to target devices/locations.

- Includes installation of physical hardware, cloud configuration, software deployment, and end-user training.
 - 6. **Maintenance phase:**
 - **Bug fixing:** Reacting to issues/defects reported by users.
 - **Performance optimisation:** Streamlining existing systems.
 - **Security updates:** Applying patches for new threats.
 - **New feature implementation:** Adding features based on changing business needs.
-

15.3 Agile approach

15.3.1 Key features and concepts of Agile

- **Definition/Core Concept:** A modern approach focused on small but continuous and rapid improvements to a system.
- **Governing Principles:**
 - **Iterative:** The project continually reviews and improves features and quality.
 - **Incremental:** Small, measurable developments and changes that move the project closer to completion.
- **Benefits:**
 - **Faster time to market:** Delivers working systems sooner with reduced initial features.
 - **Increased flexibility:** Quicker adaptation to changing requirements.
 - **Improved product quality:** Continuous feedback and testing lead to higher-quality results.
 - **Increased team morale:** Encourages teamwork and collaboration.
- **Challenges:**
 - Requires a culture change (transitioning from traditional plan-driven approaches).
 - Uncertain timelines due to feedback-dependent iterations.
 - Relies heavily on strong communication and collaboration.

15.3.2 Phases and activities of the Agile approach

- **The Sprint:** A single iteration and incrementation cycle, typically lasting between 1 and 4 weeks.
- **Phase Mechanisms (within each iteration):**
 1. **Requirements:** Identifying specific requirements for the current iteration.
 2. **Plan:** Devising an overarching but less detailed plan than Waterfall.
 3. **Design:** Using "user-centred" principles; designs are created only for the next iteration ("just-in-time" design).
 4. **Develop:** Focusing on small incremental changes identified in the plan.
 5. **Release:** New iteration released to users, monitored, and reviewed.

- [Diagram: The stages of an agile approach showing overlapping circular "Sprints" (Sprint 1, Sprint 2, Sprint 3). Each circle contains: Plan -> Design -> Develop -> Launch -> Test -> Review -> Release].

15.3.3 Agile Frameworks (Key Activities)

- **Scrum:** A popular framework involving four key activities:
 - **Sprint planning:** Defining goals and tasks for the upcoming sprint.
 - **Daily scrum:** Short daily meetings to share progress and resolve problems.
 - **Sprint review:** Collecting stakeholder feedback at the end of a sprint.
 - **Sprint retrospective:** Reflecting on the past sprint to identify future improvements.
 - **Other Frameworks:**
 - **Kanban:** Focuses on continuous flow and visualising tasks on a board.
 - **Lean:** Adapts principles from lean manufacturing to IT development.
-

15.4 Diagrams used in system development

15.4.1 Information flow diagrams (IFDs)

- **Definition/Core Concept:** Visual representations showing how information travels within and between systems, organisations, and processes.
 - **Components:**
 - **Entities:** Sources or destinations of information (customers, databases, external systems). Represented by rectangles.
 - **Flows:** Direction of information flow, shown using arrows.
 - **Processes:** Transformations or processing of information. Represented by boxes.
 - **Decisions:** Logical decisions where flow depends on specific conditions. Represented by diamonds.
 - **Levels of IFD:**
 - **Context or Level 0 diagram:** High-level overview of the entire system and external interactions.
 - **Level 1 diagram:** More detailed view focusing on major subsystems and flows within them.
 - **Level 2 diagram:** Highly detailed view exploring specific processes and handling of information.
 - [Diagram: Information flow diagram for a smart home showing a "Wireless access point" as a central hub connecting "User" (smartphone app), "Cloud-based server", "Smart thermostat", "Door lock", "Security camera", "Lights", and "Motion sensors"].
-

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Risk Score Calculation:** $RiskScore = P \times I$ Where: $P = Probability$ (1 to 5 scale)
 $I = Impact$ (1 to 5 scale)
 - **Project Scheduling Equations:** $EF = ES + D$ $LF = LS + D$ $Slack = LS - ES$
Where: $ES = EarlyStart$ $EF = EarlyFinish$ $LS = LateStart$
 $LF = LateFinish$ $D = Duration$
 - **Zero-Slack Condition:** $Slack = 0 \implies Task \in CriticalPath$
-

ULTRA-CONDENSED FINAL SUMMARY

- **Project Management:** Involves managing resources, risks (e.g., **feature creep**), and stakeholders through the **project lifecycle**.
- **Success Metrics:** Defined by **SMART targets**, clear communication, and risk management ($P \times I$).
- **Requirements: Functional** (tasks system performs) vs. **Non-functional** (performance, security, usability).
- **CPA:** Identifies the longest sequence of dependent tasks; tasks with zero **slack** are on the critical path.
- **Waterfall Method:** A linear, sequential approach (**Requirements -> Design -> Implementation -> Testing -> Deployment -> Maintenance**).
- **Agile Method:** An **iterative** and **incremental** approach using **Sprints** (e.g., Scrum framework) for faster, flexible delivery.
- **IFDs:** Use entities, flows, processes, and decisions to map data travel at three levels (0, 1, and 2).

TOPIC 16: EMERGING TECHNOLOGIES

16.1 MACHINE LEARNING

16.1.1 Concept, features and functions of machine learning

- **Definition / Core Concept:** A branch of **artificial intelligence (AI)** that allows computers to learn and improve without being explicitly programmed.
- **Governing Principle:** Machine learning (*ML*) differs from expert systems by identifying trends and correlations in large datasets and applying what it has "learned" rather than following pre-determined, fixed rules.
- **Conceptual and Mechanistic Explanation (How it works):**
 - **Algorithms:** Sets of instructions used to process data and make decisions. AI models use mathematical and statistical processes to improve performance over time.
 - **Data:** Large amounts of generated data are required for growth and accuracy. Quality and quantity (*Volume* and *Veracity*) significantly impact the learning process.
- **Types of Problems solved by *ML*:**
 - **Regression problems:** A type of **supervised learning** used to predict continuous outcomes (e.g., establishing linear relationships between variables to predict future sales based on historical data).
 - **Classification problems:** A type of **supervised learning** where the goal is to predict categorical class labels based on past observations (e.g., identifying spam emails by comparing them to known labelled "spam").
 - **Clustering problems:** An **unsupervised learning** technique used to group similar data points based on certain characteristics (e.g., grouping customers by purchase history for targeted marketing).
 - **Anomaly detection:** Used in both supervised and unsupervised models to identify data points falling outside "normal" behavior (e.g., identifying fraudulent banking transactions).
- **Training and Evaluation Mechanism:**
 - **Training set:** The larger portion of data used to train the model. The model learns patterns and adjusts internal parameters to reduce error rates.
 - **Test set:** A smaller portion of data the model is not exposed to during training. Used to evaluate quality and accuracy.
 - **Predictions:** Once the model is confirmed as working, it is applied to new, unseen data (e.g., identifying cars/pedestrians in self-driving cars).
- **Supervised vs. Unsupervised Learning:**
 - **Supervised learning:** Uses a "labelled" dataset where each training example contains an input and a corresponding output (actual category). Goal: determine the relationship (mapping function) between the data and the label.

- **Unsupervised learning:** Uses "unlabelled" or "unknown" data. The model finds hidden patterns or structures without prior knowledge or influence from bias. Used for clustering or dimensionality reduction.
- **Practical Significance:** Provides a performative improvement or competitive edge in healthcare, finance, manufacturing, and transportation.

16.1.2 Impact and possibilities of machine learning

- **Natural language processing (NLP):**
 - **Core Concept:** Focuses on interaction between computers and humans using regular everyday language.
 - **Applications:**
 - Improved search engine results (understanding user intent).
 - Virtual assistants (Siri, Amazon Alexa) paired with speech recognition to understand accents and regional variations.
 - Automated language translation (high-quality results compared to literal text translation).
 - Chatbots for customer service (improving responsiveness and efficiency).
 - **Speech recognition:**
 - **Evolution:** Early versions required training by each user and were inaccurate. Modern systems use *ML* and context to improve the quality of how spoken language is processed.
 - **Significance:** Serves as an accessibility tool or for limited-function systems (telephone helplines). Uses speech synthesis to provide distraction-free auditory feedback (e.g., while driving).
 - **Image recognition:**
 - **Core Concept:** Harnesses *ML* to identify objects within images, which is computationally difficult due to significant variations.
 - **Applications:**
 - Self-driving cars (identifying pedestrians, vehicles, traffic signs).
 - Medical imaging (identifying abnormalities in scans/X-rays for earlier intervention).
 - Security and surveillance (identifying suspicious activity or individuals).
 - Social media (automatic tagging of photos).
 - Product search (searching by image).
 - Assistive technology (identifying objects for people with sight-loss).
 - **Pattern recognition:** Finding trends, correlations, and similarities in datasets to discover and adapt to patterns in data beyond human capabilities.
-

16.2 VIRTUAL AND AUGMENTED REALITY

- **Definition / Core Concept: Virtual reality (VR)** provides an immersive experience within a computer-generated environment to create the feeling that the user is actually present.
 - **Mechanistic Components:**
 - **Headsets:** Goggle-like devices using **stereoscopic** screens (combining two images taken at slightly different angles) to provide a three-dimensional view and a sense of depth. Often include headphones and head-tracking technology.
 - **Software:** Specialized programs that generate visuals, sounds, and **haptic** feedback (simulated touch) to enhance immersion.
 - **Controllers:** Handheld devices used to interact with the virtual world and track user movements.
 - **Significance:** Surges in personal entertainment and training applications (e.g., medical simulations for doctors).
-

16.3 INTERNET OF THINGS

16.3.1 Definition and Core Concepts of IoT

- **Definition / Core Concept:** A network of devices not traditionally considered "smart" (e.g., household appliances) that have been enabled to generate, send, and receive data.
- **Governing Principles:**
 - **Connecting devices:** Equipping everyday objects with sensors for data collection and connectivity.
 - **Exchanging data:** Communication within private/limited networks (smartwatches to phones) or wider networks (internet).
 - **Automating tasks:** Making life easier/efficient via routine task automation (e.g., monitoring soil moisture to water plants).

16.3.2 Impact of the Internet of Things

- **Benefits:**
 - **Convenience:** Automation of needs (e.g., coffee machines communicating with alarm clocks).
 - **Efficiency:** Saving individuals and organisations time, energy, and resources.
 - **Safety:** Remote monitoring of hazardous systems or household sensors (e.g., carbon monoxide detectors).
- **General Impact:** Allows tracking of order progress from dispatch to delivery in real-time.

16.3.3 Challenges and Issues of the IoT

- **Standardisation:**

- **Constraint:** Different devices/systems must communicate seamlessly.
- **Mechanism:** Use of **application programming interfaces (APIs)**—sets of rules and protocols that tell systems how to communicate and share data.
- **Environmental concerns:**
 - **e-waste:** Discarded electrical equipment (computers, smartphones) may contain toxic substances. Laws now restrict disposal methods.
 - **Energy consumption:** Constant power needs for devices and the cloud infrastructure supporting them.
 - **Resource extraction:** Manufacturing requires raw materials and rare earth metals, leading to resource depletion and environmental damage. Production involving harsh chemicals contributes to greenhouse gas emissions and pollution.
- **Social/Ethical:** Some minerals come from conflict-ridden regions, raising ethical sourcing and human rights concerns.

16.3.4 Infrastructure for the Internet of Things (IoT)

- **Sensors:** Devices that monitor conditions and convert them into digital signals. They are the foundational component allowing objects to be "smart".
 - *Examples:* GPS receiver, thermistor, accelerometer, gyroscope, fingerprint sensor, microphone.
- **Networks:** Reliant on high-performance networks like *4G LTE* and *5G* to facilitate data transmission between sensors and components.
- **Embedded systems:** Specialized, integrated computing devices (microprocessor, memory, *I/O*) designed for narrow, specific tasks (e.g., washing machine, fitness tracker). They are compact, power-efficient, and reliable.
- **Storage:**
 - **Edge storage:** Data processed/stored on the device itself or nearby gateways to reduce network congestion.
 - **Cloud storage:** Centralized platforms offering scalability and accessibility.
 - **Fog computing:** Data stored in distributed mini-data centres geographically closer to devices.
 - **Hybrid storage:** Combines edge, cloud, and fog to balance performance, security, and cost.

16.3.5 Security issues relating to the IoT

- **Privacy concerns:** Collection of personal data (location, health metrics, behavior). Unauthorised access leads to identity theft and personal safety risks.
 - *Mitigation:* Strong encryption and access controls to ensure **data privacy** (data seen only by those with a valid need).
- **Data integrity:** Tampering with IoT data can have catastrophic effects (e.g., false readings in industrial systems).
 - *Mitigation:* Data validation to ensure data is within expected parameters.

- **Unauthorised access: Threat actors** (malicious/cyberthreat actors) exploit vulnerabilities (default passwords, unpatched software) for data theft or espionage.
- **Denial of service (DoS) attacks:** Overwhelming networks/devices with traffic to disrupt operations. IoT systems are particularly vulnerable due to reliance on communication channels.
- **Insecure communication channels:** Unencrypted public Wi-Fi or Bluetooth allows eavesdropping.
 - *Mitigation:* Use of encryption protocols like SSL/TLS or $IPsec$.

16.3.6 Designing an IoT solution

- **Process Requirements:** Consider objectives, individual components/sensors, and the information flow.
 - **Documentation:** Requires a written plan and an information flow diagram (IFD).
 - **Example: Smart home:**
 - *Components:* Smart thermostat, door locks, security cameras, lights, motion sensors, central hub.
 - [Diagram: Information flow diagram (IFD) for a smart home. Center: Wireless access point connected to Central hub, Cloud-based server, and User (Smartphone app). Peripherals: Door lock, Smart thermostat, Security camera, Lights, Motion sensors. Arrows show control signals from hub/app to devices and live/historic data feeds from devices to hub/app].
-

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Machine Learning Problem Types:**
 - $Regression \in Supervised Learning$ (Continuous outcomes)
 - $Classification \in Supervised Learning$ (Categorical labels)
 - $Clustering \in Unsupervised Learning$ (Grouping data)
- **Dataset Partitioning:**
 - $Total\ Data = Training\ Set + Test\ Set$
- **Network Standards:**
 - $4G\ LTE$ and $5G$
- **Infrastructure Terms:**
 - API (Application Programming Interface)
 - SSL/TLS and $IPsec$ (Security protocols)
- **Iris Data Set Parameters (Case Study):**
 - $Sepal\ length\ (cm)$ (Range: 4.30 to 7.9, Mean: 5.84, SD : 0.83, Class correlation: 0.7826)
 - $Sepal\ width\ (cm)$ (Mean: 3.02, Class correlation: -0.4194)

- *Petal length (cm)* (Mean: 3.76, Class correlation: 0.949 - High!)
 - *Petal width (cm)* (Mean: 1.20, Class correlation: 0.9565 - High!)
-

ULTRA-CONDENSED FINAL SUMMARY

- *ML*: Branch of *AI*; learns from data via algorithms (Regression, Classification, Clustering) and training/test sets. Supervised (labelled) vs. Unsupervised (unlabelled).
- *ML Impacts*: enables *NLP*, speech/image recognition (medical scans, self-driving cars), and pattern detection.
- *VR*: Immersive computer-generated environments using stereoscopic headsets, software, and haptic feedback.
- *IoT*: Network of traditionally "dumb" objects enabled to exchange data. Benefits: convenience, efficiency. Challenges: Standardisation (*API*), Security, Privacy, and Environment (**e-waste**, energy).
- *IoT Infrastructure*: Sensors (GPS, etc.), *4G/5G* networks, Embedded systems, and Storage (Edge, Cloud, Fog, Hybrid).
- **Security**: Threats include **threat actors**, *DoS* attacks, and insecure channels. Requires encryption (*SSL/TLS*, *IPsec*) and data validation for integrity.
- **Design**: Requires a written plan and an *IFD* mapping entities and data flows.